

Single Degree of Freedom Helicopter Model: Laboratory Setup Design

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Abstract—This paper presents the design of a single degree of freedom (1-DOF) Helicopter model mounted on light-weighted hollow metal rod passed through a rectangular metal block fixed between vertical support. The light weighted metal rod through the metal rod is placed such that the length from the block to its end is adjustable. A Brush Less Direct Current (BLDC) Motor, along with the Arduino controller, is connected at the end of the rod with a propeller fixed on it.

Index Terms—One Degree of Freedom Helicopter(1 DoF), PID Control Scheme, Laboratory Setup

I. INTRODUCTION

The data relevant to it is surveyed in order to understand the functioning and control aspect of the previously developed models [1], [2], [3] and [4] such as the 3 DOF Helicopter Model and 2 DOF Helicopter Model and other Control System Lab Setups like the Industrial Plant Emulator, Control Moment Gyroscope, Inverted Pendulum, MagLev Apparatus, Rectilinear Plant, Torsional Apparatus, etc.

Various other alternative controls, for different types of helicopter models have been developed previously, and with adaptable mass distribution for the helicopter flight to control the unwanted torques for the sake of smooth and comfortable flight. This is exercised using the requisite robust control algorithm based on the super twisting [5]. A similar control algorithm with linear modeling about the equilibrium point about the pitch angle, using the DK iteration method, in an unstable nonlinear the MIMO 2-DOF Helicopter system is observed [6]. In [7] the model is focused on the rejection of disturbances by sliding mode control (SMC), removing the chattering phenomenon, and fuzzy logic tunes have been established. Moreover, a comparison between SMC control and PID based on the linear quadratic regulation algorithm (LQR-PID) is implemented. A similar kind of 3-DOF helicopter with continuous sliding-mode controllers with the desired sliding dynamics using the generalized formula by Ackermann and Utkin is performed in a complete state concurrently, and high order sliding-mode differentiator is also implemented [8].

The dynamics in the helicopter model are established, and an adaptive integral backstepping algorithm is applied to achieve robust control of a 3-DOF Helicopter [9]. An investigation for the proper tracking of control problems of the elevation and travel angles by combination mixed control algorithms like normal PID [10], linear quadratic regulator (LQR) and PID-type iterative learning is proposed for

higher tracking accuracy [11]. Active disturbance rejection control(ARDC) and PID based control for the inner control loop and the outer control loop respectively is proposed for position and attitude control of small unmanned quad-rotor helicopter model and an extended state observer(ESO) is designed to observe and compensate the inner and outer uncertainties in real-time [12] [13] [14]. Further advancements in the dynamics and control of the 2-DOF helicopter include Neural Network (NN) controlling for nonlinear flight control. The neural network is apparent compared to the conventional linear quadratic regulator(LQR) [15]. Another combination in achieving a robust control in the 2-DOF helicopter system used is the Single Input Single Output (SISO) PID control along with an Uncoupled MIMO Cross-Control and Servo-tracking control with extended state observer(ESO) [16].

Fuzzy controller among the 1-DOF helicopters is also famous for regulation of the attitude of the 1-DOF helicopters [17]. Further enhancement in the dynamics of the system, where a dual rotor helicopter is proposed with a protect frame and contact force control [18]. A Model of 2-DOF helicopter with the equations of movement of Euler-Lagrange is obtained at an operating point [19]. A laboratory model of 1-DOF helicopter for exercising mechatronic systems for educational purposes is set up [20]. In [21], a generic technology underlying the novel hybrid multi-degree-of-freedom motor with improved tilting torque, practicality, and control for applications which require actuation for greater than 2-DOF is implemented. And a 2-DOF PID control with fractional order PID (FOPID) controller of Twin Rotor MIMO System (TRMS) been proposed in [22]is implemented. The major contribution and arrangement of the paper are,

- 1) Economical, frugal laboratory setup design has been showcased in chapter 2
- 2) Flexibility with the controller. In this paper Arduino Uno which has a MATLAB support package
- 3) Efficient engagement of students in teaching-learning process by implementing a closed and open-loop system in chapter 3, followed by conclusion in chapter 4

II. SYSTEM DESIGN AND DYNAMICS

A. System Design

The implementation of 1 DOF Helicopter plant is proposed such that the Helicopter Setup fixed on the metal rod suspended between a support of two vertical rigid plates, allowing

the rod to freely pitch about the horizontal axis, passing between the vertical support plates through the metal block. Setup is set, such as to move the helicopters rotor setup only along a single axis of rotation i.e. pitching angle of rotation. The complete structure is shown in Fig. 11

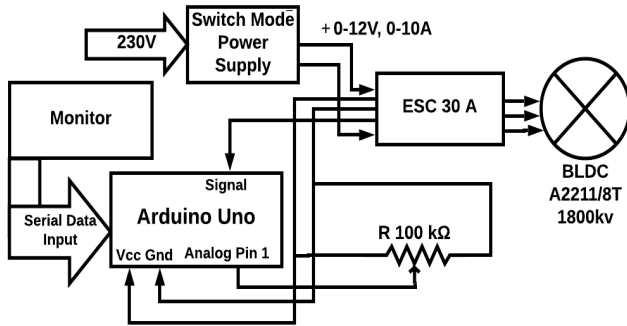


Fig. 1: 1 DOF Helicopter System Block Diagram

Shown in Fig. 1 is a Functional Block Diagram Lab setup, which altogether has five functional blocks, namely, The switch mode power supply (SMPS), a 100 kΩ Potentiometer, Arduino Uno, the Monitor, and a BLDC Motor. Each block do function differently but equally contribute in making up a complete system of 1 DOF Helicopter Model.

Three-phase Electronic Speed Controller (ESC), which is capable of driving the motors which consume up to 30A current. The best part of it is that it has a Battery Eliminator Circuit(BEC) which, as the name suggests, eliminates the need for a separate battery to power the 5V electronic device. The 100 kΩ potentiometer here plays a vital role in getting the exact value of the rod position or the travel angles with minimum error in angle positioning of the rod as it gives the readings in terms of analog data, which the controller processes the data in the range as 0 to 1023 which is further scaled to 0 to 180 degree using the *map()* function in Arduino.

B. Mechanical Modelling

1 DOF Helicopter is a model that simply consists of a base, and two vertically placed support plates with 550mm distance in between. A metal block, which is a 50mm × 40mm rectangular block(Fig. 2 and 3), is freely suspended on two cylindrical extrusions or rollers on the inner side of the vertical support plates, for which a 0.6 mm drill is made horizontally through the block. The dimensions of the metal block are shown in Fig. 2.

The two cylindrical blocks are designed differently purposely to keep the Metal Block freely suspended for its requisite movement. And an arrangement has been done in the left-hand side (LHS) cylindrical block for fixing the 100kΩ potentiometer on it. The left-hand side cylindrical block is comparatively 10mm larger in length as per the design constraints of the system. Also, the LHS cylindrical block has an indentation of 6mm diameter and length 10mm in order

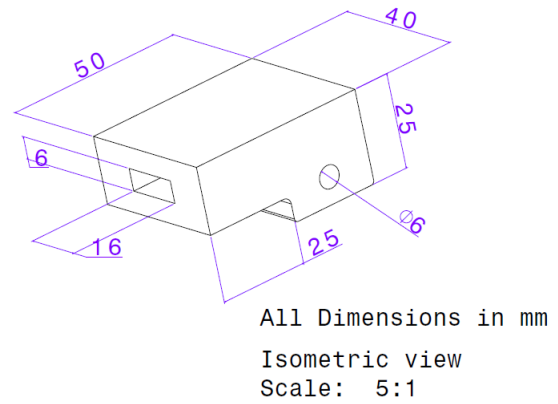


Fig. 2: Metal Block

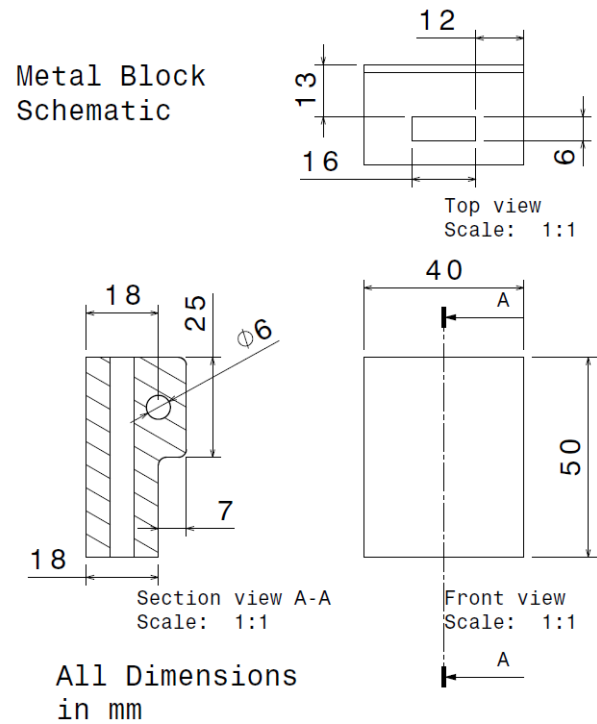


Fig. 3: Metal Block Schematic

to fix the potentiometer in it. Fig. 4 and Fig. 5 clearly show the difference between the LHS and RHS cylindrical block. Moreover, the fixing of a potentiometer with LHS cylindrical block can be seen in the Fig.6.

The metal block is specially designed to load and unload on the system easily. The Rectangular hole throughout the length of the block is specifically intended explicitly for holding the flat rod tight while the free play of the metal block when the system is in function. The cylindrical extrusions are put together with the ball-bearings to have smooth rotation. The small cylindrical extruded part is of 10mm, in which the metal block is freely suspended between the two of them, and it is

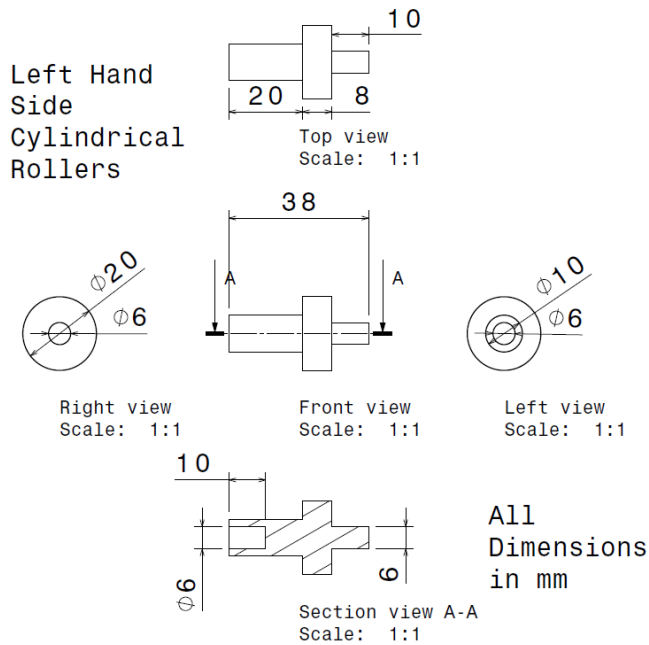


Fig. 4: LHS Cylindrical Rollers

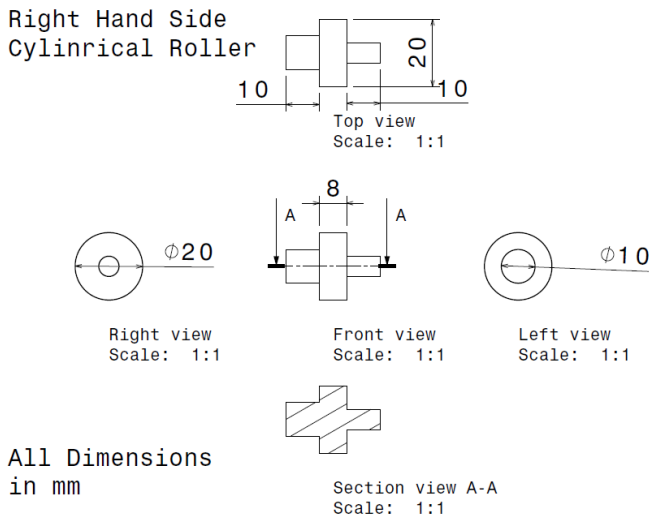


Fig. 5: RHS Cylindrical Rollers

fixed with a earing along with vertical supports. Fig. 6 clearly illustrates how the system appears when the peripherals are loaded.

A flat metal rod is passed through the rectangular pocket made throughout the length of a Metal block. The rectangular flat metal rod is a 600mm rod has made to pass through the freely suspended Metal block. The Flat rectangular metal rod, from the halfway has 0.8mm drills made after every 20mm of distance. Then the flat rod is riveted to the metal block using 0.7mm nut and bolt, which prevents the flat metal rod from slipping off during the free play of the helicopter. The flat metal rod is restricted to move only in a single degree

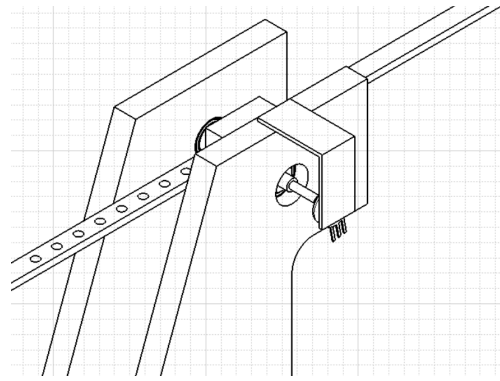


Fig. 6: Potentiometer and Cylindrical Extrusion Fixing

of freedom about the cylindrical extruded metal part. i.e., the metal rod is restricted to move only about its pitching axis. The Metal Rod is a light weighted aluminum rod precisely cut in the dimensions of 9mm by 16mm and about 610mm long. The Design for a light weighted metal rod, as shown in the Fig. 7. The base of the overall system is just a rectangular plate

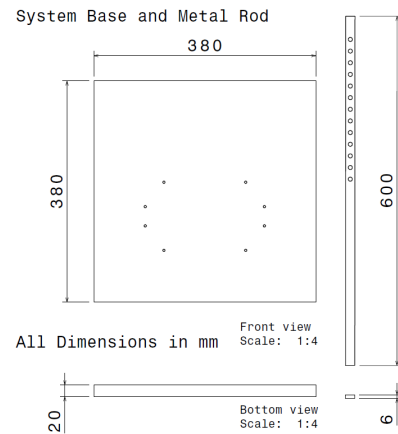


Fig. 7: System Base and Metal Rod

of 380mm x 380mm, and a thickness of 20mm provided the holes for fixing of the vertical support plate of the system. The surface area of the base is kept bigger in order to get the unnecessary subtle vibrations absorbed down or get damped during systems free play. The Design of the base is as shown in the Fig. 7.

Secondly, the vertical support plates (Fig. 8b) are designed specifically to sustain the physical load and the shocks of a system when the metal block and the rod is under free play. Also, it is designed in order to damp the unnecessary vibrations, generated form the rotary motion of the BLDC motor and the propellers, during the functioning of the system. The two vertical support plates are of 20mm thickness and have a circular indentation of about 30mm diameter for both LHS and RHS vertical plate. But the only difference both plates is the LHS has the circular indentation through out the plate as shown in Fig. 6 and Fig. 9, whereas the RHS vertical

plate has the circular indentation only halfway in the plate (i.e. of 10mm of depth), which is intentionally done to fix the potentiometer with LHS Cylindrical Block along with ball bearings to get a smooth rotary motion for the movement of the Helicopter Rod setup. The Vertical Support plates are fixed using four L-shaped rods of about 290mm long, two for each vertical plate to fix tight on the plate. The L-shaped Rod used for setting the support plates is shown in Fig. 8a.

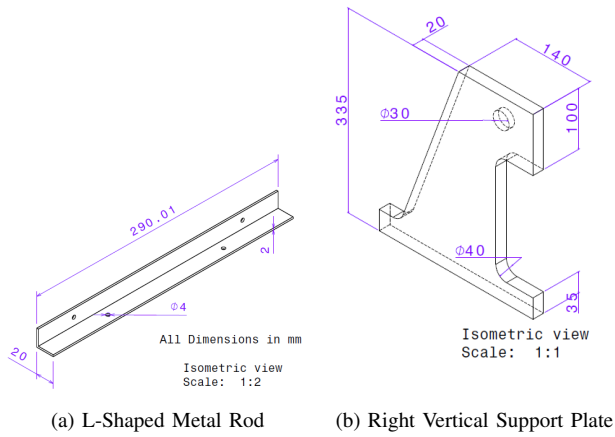


Fig. 8: Support Structure

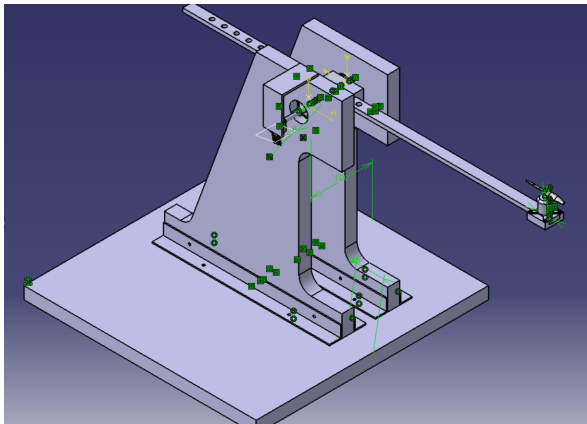


Fig. 9: System Assembly Isometric View.

C. System Operation

Figure 10 gives the complete assembly of system, with the controller board. Initially, the Arduino Uno board equipped ATmega328p is connected to the PC on which the code is dumped into the board through Arduino IDE. After all the peripherals are connected, SMPS is connected with a power supply and the system is let free for its functioning.

III. EXPERIMENTAL VERIFICATION

This chapter consists of two different phases of Implementations,

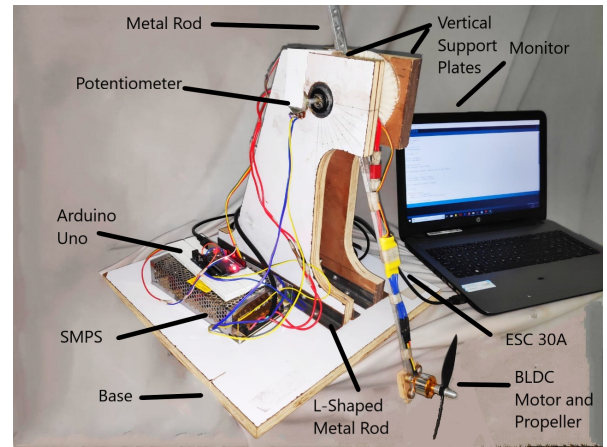


Fig. 10: Actual System Image

A. Open Loop Control of 1 DOF Helicopter

The Open-Loop Control of the 1 DOF Helicopter system is implemented using only the analog values obtained from the 100k Ohm Potentiometer. The Potentiometer is fixed along with the cylindrical block, imitates the rotary motion, and monitors the rotation of the metal block, which is under free play along pitching axis. As the Analog range for the Arduino is from 0-1023 every minute, change in the rotation of the rod about its pitching axis is detected with minute details. Further, for convenience, the Analog range of 0-1023 of the system can be converted into the equally resembling values of the angle from 0-180 degrees. During the open-loop operation of system, it is simply operated using the if-else conditional loop. The BLDC Motor is initially set in rotary motion, by setting the PWM values(which range from 0 255 in Arduino) run incrementally. The helicopter is observed to be resting for the analog values from 0 till 49, and later the motor takes a pick up in speed as the PWM value goes in crescendo in the loop. But a lift is observed at 50 point above the of analog value and is observed to be at exactly 90 degrees at the analog point of 230 (of the analog range 0-1023). At this stage, a logical conditional loop is applied that as the Helicopter rod reaches the set point of 200, the PWM count is set to reverse, and a simultaneous and gradual drop in observed in the BLDC motor speed. But as soon as the Helicopter rod lowers, a quick rise in PWM count is set so that the motor gains the required speed to reach the 230 (i.e., the 90 degrees) set point. The system behavior under such a open-loop condition does not maintain the requisite stability in the system. The system behavior under such an operational open loop can be understood from the Fig. 11

From the given Fig.11 and Fig.12, it can be concluded that the system under such an operational open loop gains a transient response if the Helicopter rod is intentionally oscillating. Which enabled such occurrence due to the absence of the feedback control.

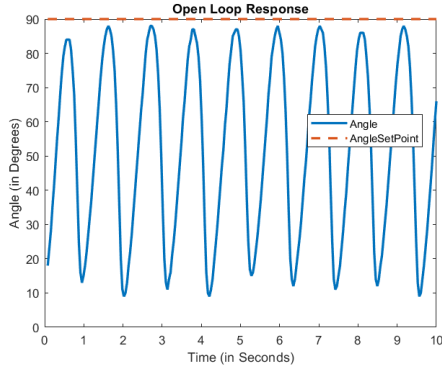


Fig. 11: Open Loop System Response (Angle)

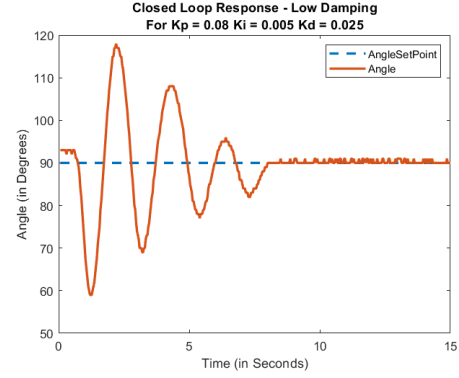


Fig. 13: Close Loop Response (Angle)

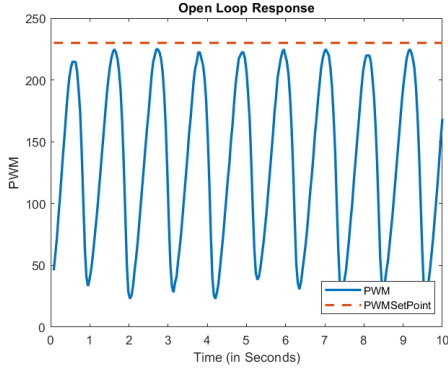


Fig. 12: Open Loop System Response (PWM)

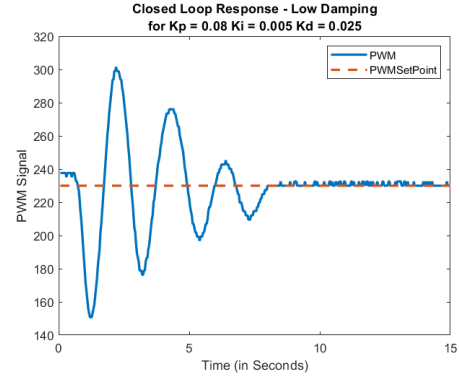


Fig. 14: Closed Loop Response (PWM)

B. Closed Loop Control of 1 DOF Helicopter

The Implementation of closed-loop control in 1-DOF Helicopter is done using the PID Algorithm in the Arduino IDE. While the implementation of the PID control in the 1 DOF Helicopter, the system functions precisely in resemblance that of the functional block diagram shown in Fig. 1. While the implementation of closed-loop control in the system, a PID library i.e., 'PID v1.h' is used. Initially, a input and a output pin for the board are assigned and then gain variables like K_p , K_i and K_d constants for proportional, integral, and derivative action respectively along with the requisite setpoint is awarded. For this particular system, the values for K_p , K_i and K_d at stability of 90 and 230 set points for angle and equivalent PWM, are observed to have values typically as 0.08, 0.05, and 0.025 respectively. At which the system relatively sustains any significant disturbance either natural or intentionally introduced to the system. The response of the helicopter rod at the gain variable points of $K_p = 0.08$, $K_i = 0.005$, $K_d = 0.025$ is observed to be, under disturbed conditions is as shown in Fig. 13 and Fig. 14 for their Angle and PWM responses receptively.

The observation for rest of the Algorithms like P , PD and PI didn't seem to be more effective in achieving the stability or the requisite output for any of the combinations of gain constants(i.e. P , PD and PI) other than PID itself for the given Angle and PWM Set points of 90 and 230 respectively.

The graph for the P appeared to be as shown in the Fig.16 given for the Proportional gain value of $K_p = 0.4$.

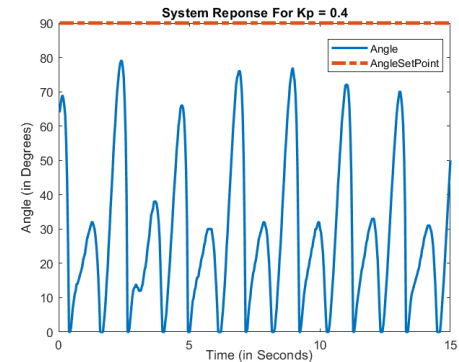


Fig. 15: System Response for $K_p = 0.4$ (Angle)

For other values of K_p above 0.5, the system behavior was unstable, and helicopter rod setup give a sudden jerk in a continuously fast way. Thus,as a resultant, makes the system unstable. But for values lower than 0.2, the motor takes more time to operate or even wont work due such a low value of proportional constant. For the PD Algorithm, the similar kind of sudden jerks are observed in the system for $K_p = 0.4$ and $K_d = 0.1$ and above the value of the same. In the case of PI, the system is noted to have a very long delay in reaching the

given Angle and PWM Set points of 90 and 230 receptively. The observation for PI is observed to obtained, as shown in Fig.16 for the gain values of $K_p = 0.5$ and $K_i = 0.005$.

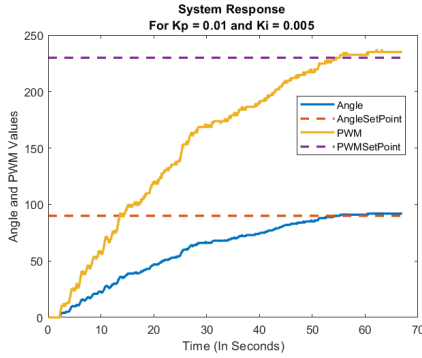


Fig. 16: System Response for $K_p = 0.1$ and $K_i = 0.005$

In the Fig. 13 and 14 given above, the PID Gain constants are $K_p = 0.08$, $K_i = 0.005$ and $K_d = 0.025$, seem to have a low damping factor. Where as changing $K_p = 0.01$ seems to have a relatively high damping factor. The System Response for PI and PID is illustrated in Fig.16 and Fig.17, respectively.

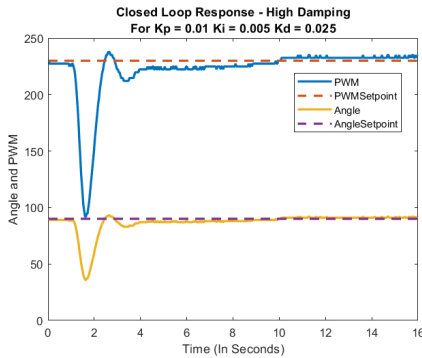


Fig. 17: System Response PID High Damping

IV. CONCLUSION

Practical and frugal design steps for designing single degree of freedom (1-DoF) laboratory test bench are proposed. The System has a simple, adaptable design, which can be assembled or disassembled easily without much effort. Also the system is portable enough and transportable. The helicopter model is mount on light-weighted metal rod passed through a rectangular metal block. The metal rod is designed such that to adjust its length and fix it in the metal block between vertical support. The performance in open-loop and closed-loop control is studied on the Arduino board. The result shows that the oscillating nature can be avoided by considering the feedback of rod position while bringing stability in the system. Scope of this lab-setup will help learners to study and validate essential components of control systems like modeling, operation of BLDC motor along with sensor interface, easy validation of open and closed-loop effect on the system.

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